

CITY UNIVERSITY - DHAKA, BANGLADESH

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Course Resource: Data Structures & Algorithms (CSE 2115)

Faculty: Department of Computer Science & Engineering

Resource Type: Lecture Notes, Syllabus Reference & Lab Exercises

Target Semester: Semester 4 | Target Batch: Batch 67 - Section A & B

1. Course Overview & Learning Objectives

This sample academic document is provided to students enrolled in City University degree programs. The course covers asymptotic analysis, abstract data types, recursive strategies, and algorithmic problem solving in modern software engineering environments.

2. Curriculum Breakdown & Topic Coverage

- * Module 1: Complexity Analysis - Big-O, Big-Omega, Big-Theta and Recurrence Relations
- * Module 2: Linear Structures - Dynamic Arrays, Singly/Doubly Linked Lists, Stacks, Queues
- * Module 3: Trees & Graphs - Binary Search Trees, AVL Self-Balancing, Heap Structures
- * Module 4: Traversal & Paths - Breadth-First Search (BFS), Depth-First Search (DFS)
- * Module 5: Greedy Algorithms & Dynamic Programming - Knapsack, Shortest Paths (Dijkstra)

3. Laboratory Assessment & Project Guidelines

Practical sessions are conducted weekly. All students must maintain documented lab reports and submit functional code solutions adhering to coding style guidelines and test coverage.

4. Academic Integrity & Institutional Policy

City University upholds the highest standards of academic honesty. Plagiarism or unauthorized redistribution of examination questions and lab manuals will incur institutional penalties.